# PRODUCT DESIGN
# Part-2

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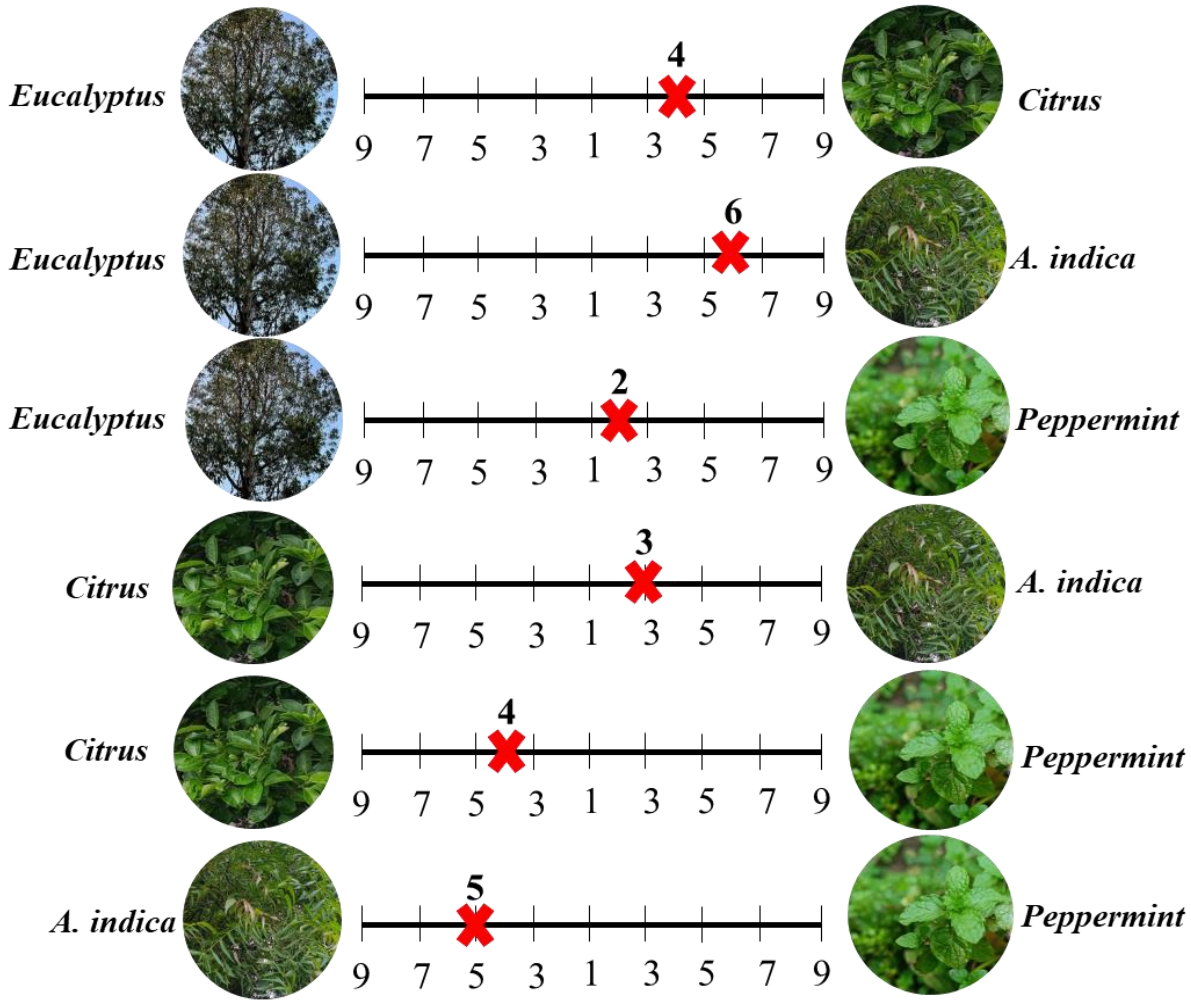
# Analytical Hierarchy Process (AHP)

- Formal method for determining relationships between discrete alternatives
- Each alternative is rated by one or more attributes

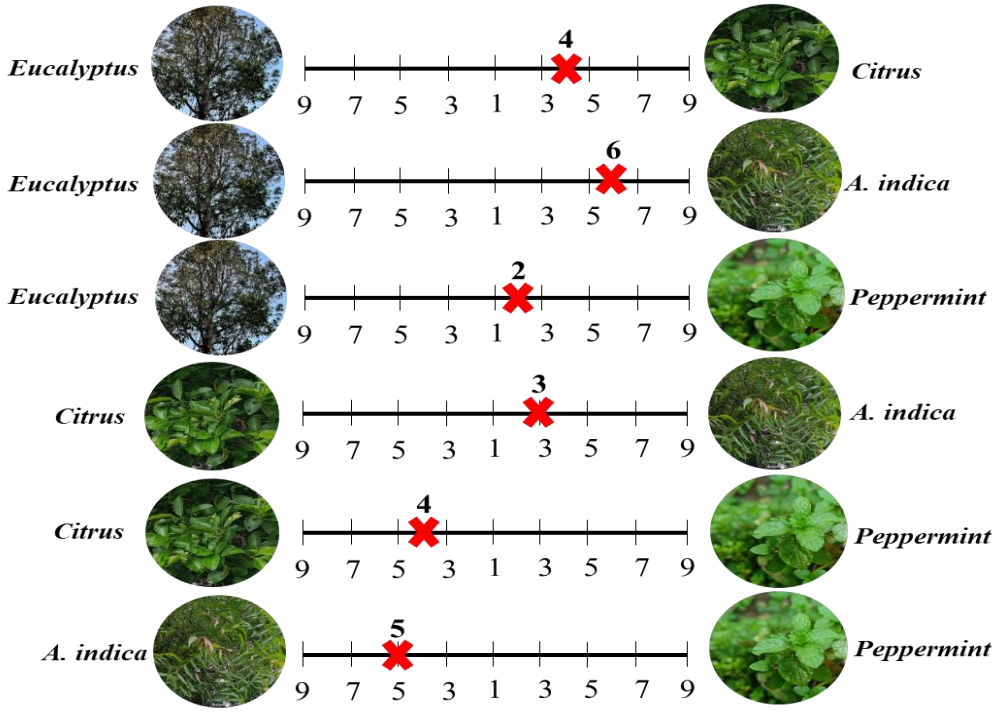
  It has an axiomatic foundation which requires that the annihilation condition is not satisfied.

## An AHP example from the following paper:

Pagidi Madhukar, Lalit M. Pandey, Uday S. Dixit,
Post-harvest grain storage: Methods, factors, and eco-friendly solutions, Food Control, Volume 174, 2025, 111236,
ISSN 0956-7135, https://doi.org/10.1016/j.foodcont.2025.111236.




Paired comparison of four different plant species on a scale of 1–9




$$C_m = \begin{array}{c} \\ Eucalyptus \\ Citrus \\ A.\ indica \\ Peppermint \end{array} \begin{bmatrix} Eucalyptus & Citrus & A.\ indica & Peppermint \\ 1 & \frac{1}{4} & \frac{1}{6} & \frac{1}{2} \\ 4 & 1 & \frac{1}{3} & 4 \\ 6 & 3 & 1 & 5 \\ 2 & \frac{1}{4} & \frac{1}{5} & 1 \end{bmatrix}$$

$$C_m = \begin{array}{c} \\ Eucalyptus \\ Citrus \\ A.\ indica \\ Peppermint \\ \text{SUM} \end{array} \begin{bmatrix} Eucalyptus & Citrus & A.\ indica & Peppermint \\ 1 & \frac{1}{4} & \frac{1}{6} & \frac{1}{2} \\ 4 & 1 & \frac{1}{3} & 4 \\ 6 & 3 & 1 & 5 \\ 2 & \frac{1}{4} & \frac{1}{5} & 1 \\ 13 & \frac{9}{2} & \frac{17}{10} & \frac{21}{2} \end{bmatrix}$$

$$C_m = \begin{array}{c} \\ Eucalyptus \\ Citrus \\ A.\ indica \\ Peppermint \\ \text{SUM} \end{array} \begin{bmatrix} Eucalyptus & Citrus & A.\ indica & Peppermint \\ \frac{1}{13} & \frac{1}{18} & \frac{5}{51} & \frac{1}{21} \\ \frac{4}{13} & \frac{2}{9} & \frac{10}{51} & \frac{8}{21} \\ \frac{6}{13} & \frac{2}{3} & \frac{10}{17} & \frac{10}{21} \\ \frac{2}{13} & \frac{1}{18} & \frac{2}{17} & \frac{2}{21} \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Next, the normalized principal Eigenvector was obtained by summation of the values in the rows and taking their average. It was also known as a priority vector:

$$\mathbf{W} = \frac{1}{4}\begin{bmatrix} \frac{1}{13} + \frac{1}{18} + \frac{5}{51} + \frac{1}{21} \\ \frac{4}{13} + \frac{2}{9} + \frac{10}{51} + \frac{8}{21} \\ \frac{6}{13} + \frac{2}{3} + \frac{10}{17} + \frac{10}{21} \\ \frac{2}{13} + \frac{2}{18} + \frac{2}{17} + \frac{2}{21} \end{bmatrix} = \begin{bmatrix} 0.0695 \\ 0.2767 \\ 0.5482 \\ 0.1056 \end{bmatrix}$$

$$C_m = \begin{array}{c} \\ Eucalyptus \\ Citrus \\ A.\ indica \\ Peppermint \\ \text{SUM} \end{array} \begin{array}{c} \begin{bmatrix} Eucalyptus & Citrus & A.\ indica & Peppermint \\ 1 & \frac{1}{4} & \frac{1}{6} & \frac{1}{2} \\ 4 & 1 & \frac{1}{3} & 4 \\ 6 & 3 & 1 & 5 \\ 2 & \frac{1}{4} & \frac{1}{5} & 1 \end{bmatrix} \\ \begin{array}{cccc} 13 & \frac{9}{2} & \frac{17}{10} & \frac{21}{2} \end{array} \end{array}$$

$$\mathbf{W} = \frac{1}{4}\begin{bmatrix} \frac{1}{13}+\frac{1}{18}+\frac{5}{51}+\frac{1}{21} \\ \frac{4}{13}+\frac{2}{9}+\frac{10}{51}+\frac{8}{21} \\ \frac{6}{13}+\frac{2}{3}+\frac{10}{17}+\frac{10}{21} \\ \frac{2}{13}+\frac{2}{18}+\frac{2}{17}+\frac{2}{21} \end{bmatrix} = \begin{bmatrix} 0.0695 \\ 0.2767 \\ 0.5482 \\ 0.1056 \end{bmatrix}$$

$$\lambda_{\max} = 0.0695 \times 13 + 0.2764 \times \frac{9}{2} + 0.5482 \times \frac{17}{10} + 0.1056 \times \frac{21}{2}$$
$$= 4.188$$

$$C_I = \frac{\lambda_{max} - n}{n-1}.$$

Putting $n = 4$ and $\lambda_{max} = 4.188$, consistency index $C_I$ was obtained as 0.063

Random consistency index

| n | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RCI | 0.58 | 0.90 | 1.12 | 1.24 | 1.32 | 1.41 | 1.45 | 1.49 | 1.51 | 1.48 | 1.56 | 1.57 | 1.59 |

Source: Adapted from Saaty (1980), pp. 21.

$$C_R = \frac{C_I}{R_I} = 0.07.$$

It is less than 0.1; hence, consistent.

Saaty, T.L. (1980) **The Analytic Hierarchy Process**. McGraw-Hill, New York

$$\mathbf{W} = \frac{1}{4}\begin{bmatrix} \frac{1}{13} + \frac{1}{18} + \frac{5}{51} + \frac{1}{21} \\ \frac{4}{13} + \frac{2}{9} + \frac{10}{51} + \frac{8}{21} \\ \frac{6}{13} + \frac{2}{3} + \frac{10}{17} + \frac{10}{21} \\ \frac{2}{13} + \frac{2}{18} + \frac{2}{17} + \frac{2}{21} \end{bmatrix} = \begin{bmatrix} 0.0695 \\ 0.2767 \\ 0.5482 \\ 0.1056 \end{bmatrix}$$

A. Indica corresponding to third row is the best.

# Taguchi’s method

- Design variable set is defined by the expected variation of a single performance variable from target value due to stochastic noise
- Probabilistic Optimization methods can be used to evaluate the expected value of the objective function.

# Fuzzy Design method

- Design imprecision is manipulated using mathematics of fuzzy sets.
- Imprecision is represented by a range and a function is defined over that range to describe preferences for particular values.
- Real-valued attributes as well as non-parametric attributes can be considered.

# Fuzzy set applications

- Fuzzy operations

$$A = \{0.7/\mathrm{L1},\ 0.9/\mathrm{L2},\ 0.8/\mathrm{L3}\},\quad B = \{0.8/\mathrm{L1},\ 0.5/\mathrm{L2},\ 0.6/\mathrm{L3}\}$$

- Fuzzy Arithmetic $A \cap B = \{0.7/\mathrm{L1},\ 0.5/\mathrm{L2},\ 0.6/\mathrm{L3}\}$

- Fuzzy Logic

# Applications of fuzzy set operations

- If a product has a membership grade $\mu_1$ in one objective and $\mu_2$ in other objective, its overall membership grade can be defined as

$$\mu_o = \min(\mu_1, \mu_2)$$

- Here, the overall performance of the model is dependent on the most poorly performing aspect. This type of strategy is called non-compensating strategy.

# Applications of fuzzy set operations

- An overall membership grade based on a compensating strategy may be defined

$$\mu_o = \sqrt{\mu_1 \mu_2}$$

- A weighted combination of two strategies may also be taken. Thus, if membership grades of a model in *n* different objectives are , $\mu_1, \mu_2, \ldots\ldots\ldots, \mu_n$

, then overall membership grade may be defined as

$$\mu_o = (1-\alpha)\min(\mu_1, \mu_2, ,,,,,, \mu_n) + \alpha \sqrt[n]{\mu_1 \mu_2 ,,, \mu_n}$$

# Applications of fuzzy set operations

- One can easily incorporate linguistic information in the decision process. For example, if a component has a membership grade $\mu$ in a set of 'good surface finish', it can be assigned a membership grade of $\sqrt{\mu}$ in the set of 'more or less good surface finish'. Thus, customers' requirement available in the language form may be converted into mathematical form.

$\sqrt{\mu}$

# Applications of fuzzy set operations

- In the area of multi-objective machining optimization, the fuzzy set based decision strategy may be very effective. There are a number of objectives to be satisfied in a multi-objective machining optimization such as minimum cost of product, maximum production rate, maximum profit rate and minimum surface roughness. In the fuzzy set based approach, different solutions may be assigned membership grades in different objectives and then based on the fuzzy set operations, the solution having the overall membership grade may be chosen.

# Method of Imprecision

- Constraints can be imprecise rather than crisp.
- Permits the customer to specify preference over a range of values.

# Method of Imprecision

Involves two strategies:

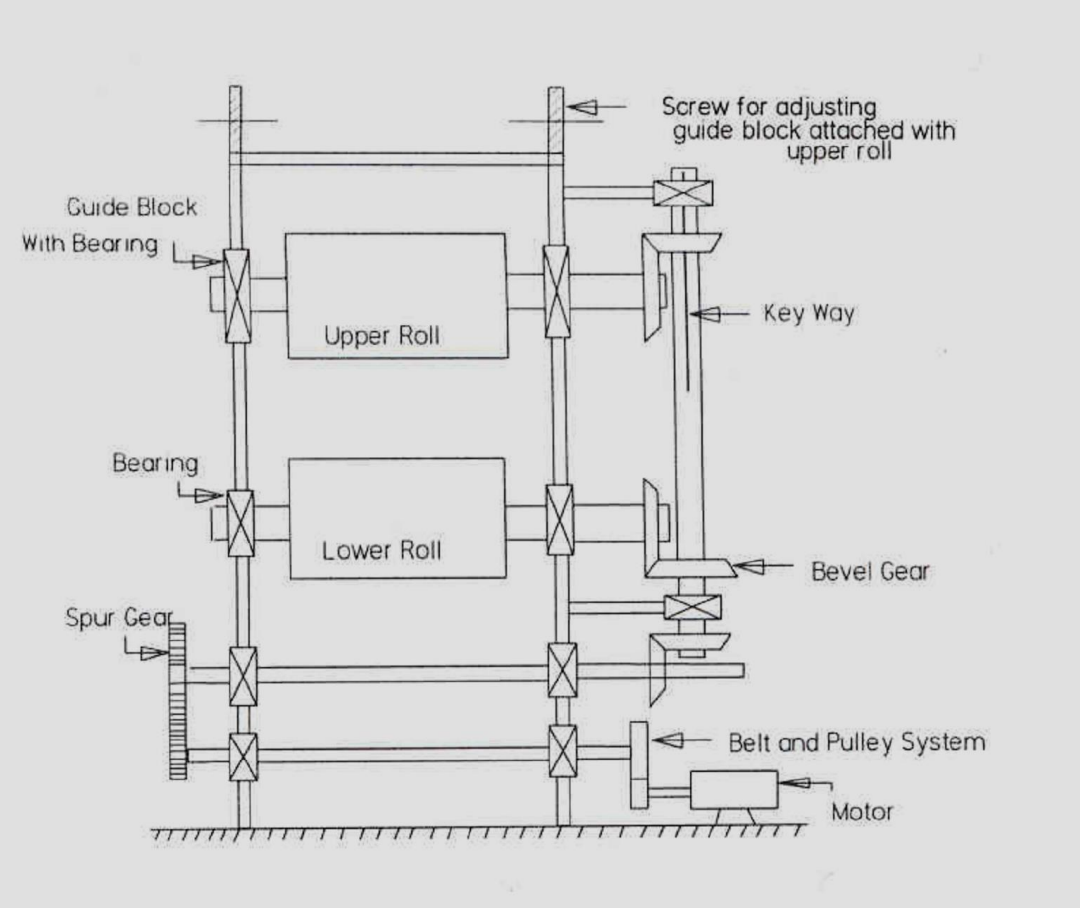
1. Compensating strategy- in which aspects performing poorly are compensated by aspects performing well.
2. Non-compensating strategy- in which poor performance in one aspect cannot be compensated by good performances in others.

# Rolling mill design

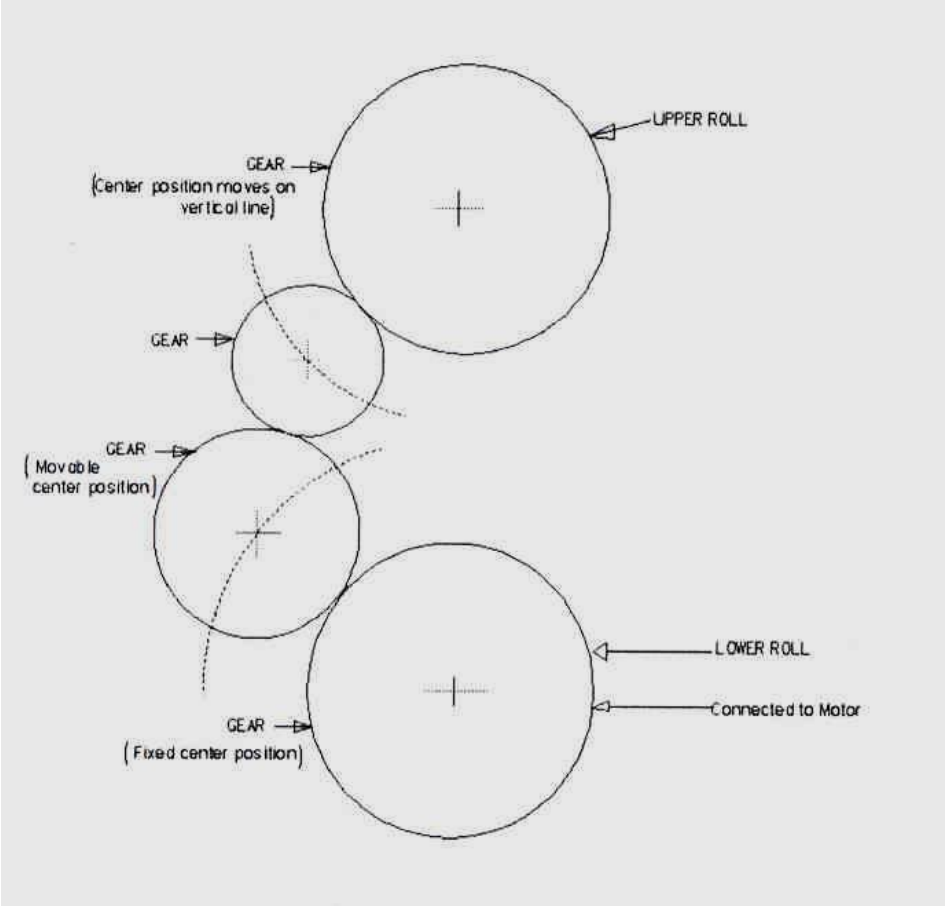
Three alternatives for roll gap adjustment:

1. Bevel gear drive
2. Spur gear drive
3. Universal joint drive

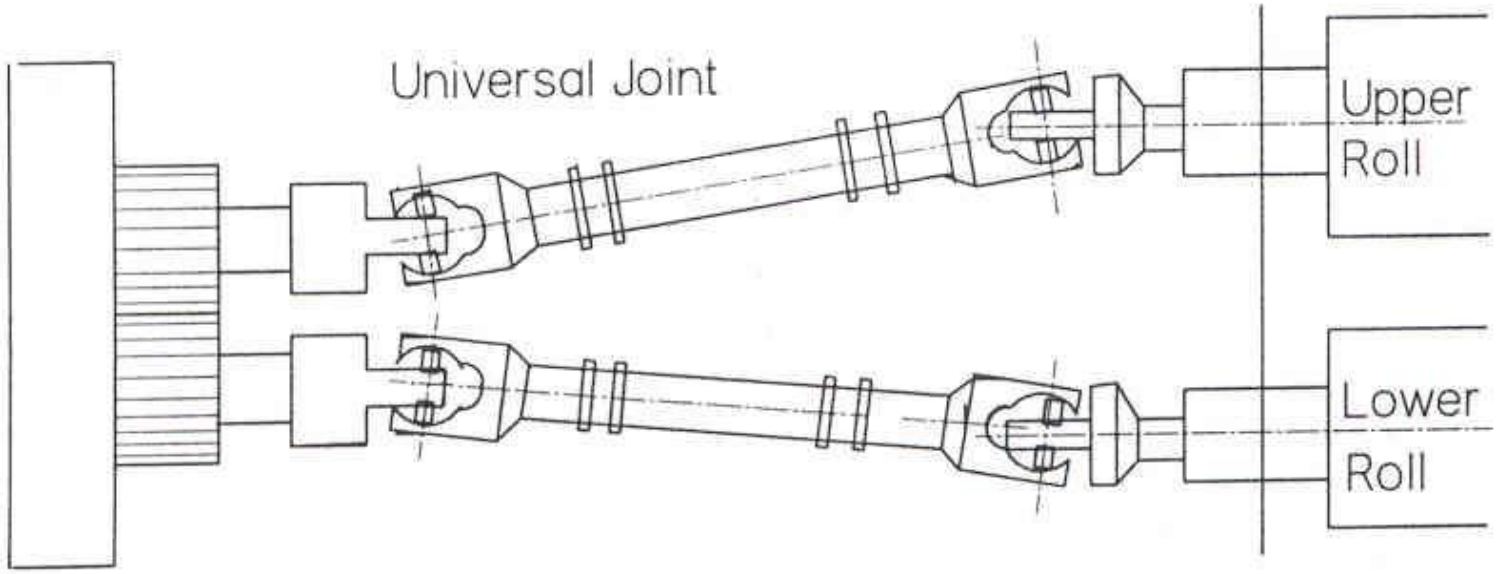
# Bevel gear drive for rolling mill



# Spur gear drive for rolling mill



# Universal joint drive for rolling mill



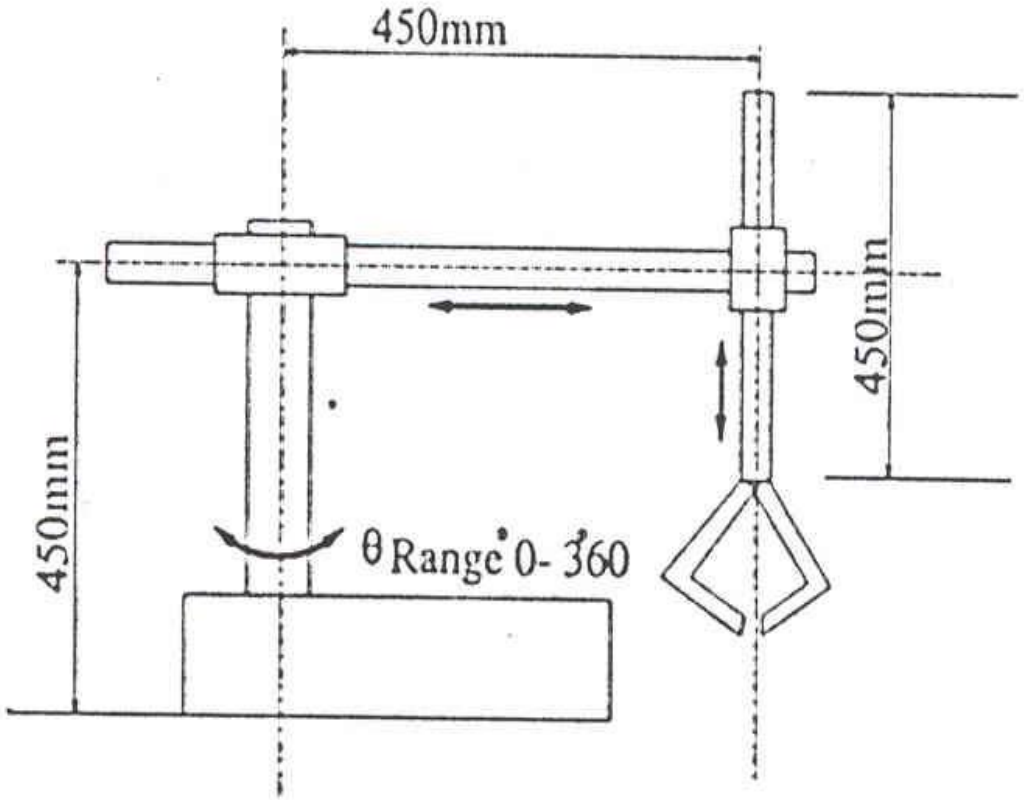
| Sl. No | Alternatives | | Bevel gear drive | | Spur gear drive | | Universal joint drive | |
|---|---|---|---|---|---|---|---|---|
| | Criteria | Importance(I) | Ratings(Ra) | Ra×I | Ratings(Ra) | Ra×I | Ratings(Ra) | Ra×I |
| 1 | Design simplicity | 10 | 7 | 70 | 7 | 70 | 10 | 100 |
| 2 | Economy | 9 | 8 | 72 | 9 | 81 | 9 | 81 |
| 3 | Ease of assembly | 8 | 6 | 48 | 6 | 48 | 8 | 64 |
| 4 | Ease of maintenance | 8 | 7 | 56 | 8 | 64 | 8 | 64 |
| 5 | Efficiency of power transmission | 9 | 8 | 72 | 8 | 72 | 8 | 72 |
| 6 | Reliability | 9 | 8 | 72 | 8 | 72 | 9 | 81 |
| | Total points | | | 390 | | 407 | | 462 |

# Robot design

Four alternatives for conversion of rotary motion to reciprocating motion:

1. Rack and pinion
2. Slider crank
3. Screw and nut
4. Cam and follower

# Robot line diagram



| Sl. No | Alternatives | | Rack and Pinion | | Slider crank | | Screw and nut | | Cam and follower | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Criteria | I | Ra | Ra×I | Ra | Ra×I | Ra | Ra×I | Ra | Ra×I |
| 1 | Design Simplicity | 5 | 8 | 40 | 5 | 25 | 10 | 50 | 4 | 20 |
| 2 | Precision | 4 | 8 | 32 | 5 | 20 | 7 | 28 | 5 | 20 |
| 3 | Resolution | 4 | 8 | 32 | 6 | 24 | 10 | 40 | 2 | 8 |
| 4 | Efficiency | 10 | 10 | 100 | 10 | 100 | 2 | 20 | 10 | 100 |
| 5 | Ease of Manufacturing | 10 | 6 | 60 | 10 | 100 | 8 | 80 | 3 | 30 |
| 6 | Ease of assembly | 10 | 8 | 80 | 8 | 80 | 8 | 80 | 8 | 80 |
| 7 | Smoothness of operation | 6 | 10 | 60 | 6 | 36 | 8 | 48 | 4 | 24 |
| | Total Points | | | 404 | | 385 | | 346 | | 282 |

PRIORITY DECISION TABLE

# Conclusions

- There are a number of ways to approach the design problem of a particular product.
- Formal design method provided a crisp method to choose among many possible designs.
- If formal method of design is employed right from early stages of design, better product design will emerge.